

A locked-protocol benchmark of LoRA-adapted AgroNT versus a DeepSEA-style CNN for tissue-averaged expression classification in *Arabidopsis thaliana*

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ABSTRACT Models that classify gene expression from flanking DNA would help plant geneticists annotate uncharacterized genes and prioritize noncoding variants. Compact convolutional networks and plant DNA foundation models are rarely scored on the same genes, labels, metrics, and stopping rules. We compared a DeepSEA-style one-dimensional convolutional network with AgroNT-1B, adapted by low-rank updates, on a tissue-averaged high-versus-low task in *Arabidopsis thaliana*. Labels average 56 Plants Genomic Benchmark tissue abundances and threshold at the training-set median. This derived binary collapse is not the official multi-tissue regression. On 3,402 held-out genes, AgroNT (seed 42) reached 86.8% accuracy and AUROC 0.939, versus 75.4% and 0.842 for the convolutional network (difference 0.097; 95% confidence interval 0.086–0.108). Three convolutional seeds gave AUROC 0.844 ± 0.003 . Single-tissue labels agreed with the averaged label for 92.1% of genes (bootstrap 95% confidence interval 0.912–0.928 across 56 tissues). Variant-effect maps were architecture-specific under the stated allele protocols. The same *Arabidopsis* checkpoints kept that ranking on rice, maize, tomato, and soybean tests; AgroNT pretraining already included those genomes, so that readout is adapter transfer rather than a species-unseen test. Under this locked protocol, low-rank-adapted AgroNT outperforms this convolutional network by a statistically supported margin. Code and processed tables are public; fine-tuned checkpoints are reproduced by retraining.

KEYWORDS
gene expression
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INTRODUCTION

Most of a plant genome is noncoding DNA. Cis-regulatory elements in promoters, 5'/3' untranslated regions (UTRs), and terminators determine when, where, and at what level genes are expressed (Marand *et al.* 2023). A sequence-to-expression scorer that reads this code from DNA would support annotation of uncharacterized genes, ranking of noncoding variants, and design of synthetic regulatory DNA (Hu *et al.* 2023).

Convolutional neural networks (CNNs) showed that proximal DNA carries substantial information about transcript abundance. DeepSEA established this for noncoding variant effects (Zhou and Troyanskaya 2015). In plants, Peleke *et al.* (2024) trained interpretable CNNs that classify high-versus-low expression from gene-flanking sequence across four species with >80% accuracy and recovered known transcription-factor

motifs by attribution. In parallel, DNA language models pretrained on whole genomes have become general-purpose genomic feature extractors (Avsec *et al.* 2021). AgroNT, a one-billion-parameter transformer pretrained on ~10.5 million sequences from 48 edible plant genomes, reported strong results across the Plants Genomic Benchmark (PGB) (Mendoza-Revilla *et al.* 2024; InstaDeepAI 2024).

What remains unclear for applied plant genetics is how large the gain of foundation-model fine-tuning is relative to a compact CNN under identical splits, labels, metrics, and early-stopping rules, and which sequence compartments carry that prediction. Here we provide that comparison in *Arabidopsis thaliana* (Figure 1). The contest is a locked-protocol gain: a DeepSEA-style 1D-CNN versus AgroNT fine-tuned with low-rank adaptation (LoRA) on the official *Arabidopsis* PGB split with tissue-averaged binary labels (Figure 2). Complementary readouts ask where predicted sensitivity sits in annotated flanks (Figure 3) and whether the *Arabidopsis* ranking holds when the same checkpoints are applied to crop PGB test sets (Figure 4). The four contributions are therefore (i) a head-to-head score under one protocol, (ii) architecture-specific regional maps reported

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with their scoring rules, (iii) the numerical scope of tissue-averaged labels, and (iv) public code and processed tables from which the checkpoints can be retrained.

4 MATERIALS AND METHODS

5 Data

We used the *Arabidopsis thaliana* gene-expression FASTA of the Plants Genomic Benchmark (Mendoza-Revilla *et al.* 2024; InstaDeepAI 2024). Each gene is represented by a 6-kb genomic window constructed as 5 kb upstream plus 1 kb downstream of the annotated transcription start site (TSS). Expression labels in the FASTA comprise 56 tissue/sample RNA-sequencing measurements. The official split comprises 25,731 training, 3,401 validation, and 3,402 test genes, with no gene overlap across splits. The official PGB gene-expression task is multi-tissue regression; here we derived binary high/low labels from the median of per-gene mean expression on the training set. For gene g with tissue abundances $y_{g,t}$ ($t = 1, \dots, 56$),

$$\bar{y}_g = \frac{1}{56} \sum_{t=1}^{56} y_{g,t}, \quad z_g = \mathbf{1}\{\bar{y}_g \geq \tau\}, \quad \tau = \text{median}\{\bar{y}_g : g \in \text{train}\}. \quad (1)$$

The training-set threshold was $\tau = 5.605$. Because labels aggregate expression across 56 tissues/samples, the benchmark targets sequence correlates of tissue-averaged abundance class, not organ-specific programs. Conclusions from this paper apply to that derived binary task and should not be read as a substitute for the official PGB multi-tissue regression.

In parallel, we built a structured gene-centric dataset for region-level interpretability. For each of 27,436 protein-coding *A. thaliana* genes with usable flanking annotation and expression (TAIR10/Araport11; Ensembl Plants release 59 (Arabidopsis Genome Initiative 2000; Lamesch *et al.* 2012; Harrison *et al.* 2024)), we extracted the promoter (1 kb upstream of the TSS), 5'UTR (500 nt), 3'UTR (500 nt), and terminator (1 kb downstream of the transcription termination site) in transcriptional orientation, concatenated the four regions 5' → 3' with a 20-nt N-padding gap, and one-hot encoded the resulting 3,020-bp sequence. Genes were assigned to fixed chromosomal splits (train: chromosome 5, $n = 6,331$; validation: chromosomes 3 and 4, $n = 9,636$; test: chromosomes 1 and 2, $n = 11,469$). Binary labels used the training-split median of $\log_2(\text{mean expression} + 1)$ (threshold 3.038).

35 Model architectures

1D-CNN baseline. We implemented a DeepSEA-style convolutional network (Zhou and Troyanskaya 2015) with three convolutional blocks, each containing two 1D convolutions (256 filters, kernel size 8), batch normalization, ReLU activation, and max pooling (pool size 4), followed by global max pooling and two fully connected layers with dropout. The model has 2.7 M parameters.

DNA foundation model. AgroNT-1B (Mendoza-Revilla *et al.* 2024) is a transformer of 985 M parameters pretrained on ~10.5 million sequences from 48 edible plant genomes. We adapted it for sequence classification with LoRA adapters (Hu *et al.* 2021) on the query/key/value/output projections of every self-attention block, plus a two-layer head over mean-pooled hidden states. For a frozen pretrained weight W_0 ,

$$W = W_0 + \frac{\alpha}{r} BA, \quad r = 16, \alpha = 32, \quad (2)$$

where B and A are the trainable low-rank factors (PEFT default scaling). Only the 16.2 M LoRA and head parameters (1.6% of the total) were trainable.

Training, interpretability, tissue labels, and variants

Both models were trained with binary cross-entropy and early stopping on validation area under the receiver operating characteristic curve (AUROC). For predicted high-expression probability p_i and label z_i ,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [z_i \log p_i + (1 - z_i) \log(1 - p_i)]. \quad (3)$$

The CNN used batch size 64 and AdamW (learning rate 10^{-3}) for up to 30 epochs. AgroNT used an effective batch size of 32 and AdamW (learning rate 2×10^{-4}) for up to 3 epochs, all on a single NVIDIA RTX 4090. Reported metrics were accuracy, AUROC, and area under the precision-recall curve (AUPRC). The PGB CNN was repeated with three seeds (42/43/44); AgroNT is reported from seed 42. An input-length ablation retrained both models after midpoint cropping of the 6 kb windows to 3 kb and 1.5 kb. Because that window is 5 kb upstream plus 1 kb downstream of the TSS, midpoint cropping removes most TSS-proximal sequence.

Matched region importance used sliding-window mutagenesis on structured flanking sequences (50-bp N substitution, 25-bp step). For sequence x and window w , a mutated copy $x_{w \leftarrow N}$ replaces the window with N bases. Position-wise importance is the mean absolute change over all windows that cover base j ,

$$I_j = \frac{1}{n} \sum_{i=1}^n \sum_{w \ni j} |p(x^{(i)}) - p(x_{w \leftarrow N}^{(i)})|, \quad (4)$$

and the region score is the mean of I_j inside that compartment. The CNN for this analysis was retrained on the concatenated flanking set (test AUROC 0.805). AgroNT mutagenesis applied the Arabidopsis PGB LoRA checkpoint to the same concatenated strings without retraining on that representation. DeepLift attributions (Shrikumar *et al.* 2017) on structured test genes supported TF-MoDisco-lite (Shrikumar *et al.* 2018) and TOMTOM annotation against JASPAR 2024 (Rauluseviciute *et al.* 2024). Architecture comparisons of region ranks use the matched mutagenesis scores rather than DeepLift magnitudes.

We parsed the 56 tissue/sample values in PGB FASTA headers for 32,534 unique Arabidopsis genes and computed pairwise Spearman correlations, per-gene cross-tissue coefficient of variation (CV), and agreement between the tissue-averaged high/low label and each tissue's own median-based label (Figure S1).

Natural single-nucleotide variants from the Ensembl Plants *A. thaliana* variation VCF (Harrison *et al.* 2024) were mapped onto the structured flanking representation of 250 genes. For reference and alternate sequences x_{ref} and x_{alt} ,

$$|\Delta| = |p(x_{\text{alt}}) - p(x_{\text{ref}})|. \quad (5)$$

The CNN scored 27,662 variants by substituting the mapped alternate allele. AgroNT scored 26,518 overlapping non-gap sites as the mean $|\Delta|$ over the three non-reference bases. Region ranks are therefore compared within each model, not as a unified allele-matched contest. An overlap of 17 reported expression quantitative trait locus (eQTL) associations was treated as exploratory (Figure S2).

Cross-species transfer on crop PGB tests

Frozen Arabidopsis CNN and AgroNT checkpoints were applied, without crop fine-tuning, to the official PGB test sets of rice (*Oryza sativa*; $n = 3,702$), maize (*Zea mays*; $n = 4,483$), tomato (*Solanum lycopersicum*; $n = 3,827$), and soybean (*Glycine max*; $n = 4,803$). Labels used each species' training-split median. These four test sets were not used to train or select the Arabidopsis models. AgroNT was pretrained on genomes that include these crops (Mendoza-Revilla *et al.* 2024), so the crop readout is adapter transfer on a multi-species backbone; the CNN never trained on crop genomes. The two readouts are therefore not interchangeable species-unseen tests.

Statistical analysis

The CNN–AgroNT difference used 500-fold bootstrap 95% confidence intervals on the test set and McNemar’s test on paired classifications. With $b = 142$ genes classified correctly only by the CNN and $c = 532$ classified correctly only by AgroNT, the continuity-corrected statistic is

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}. \quad (6)$$

Region-wise variant $|\Delta|$ comparisons used two-sided Mann–Whitney tests, Benjamini–Hochberg false-discovery rates across the six pairwise tests within each architecture, rank-biserial effect sizes, and 10,000-resample bootstrap 95% confidence intervals for regional medians. Tissue-label concordance is reported as the mean agreement across 56 tissues with a bootstrap 95% CI obtained by resampling tissues. Crop AUROC values are reported per species without pooling. The 17-variant eQTL overlap used a one-sided Mann–Whitney test against the CNN background and a 10,000-iteration permutation test of the eQTL median; both are exploratory.

Data availability

No unique biological reagents were generated. Primary inputs are public: PGB Arabidopsis and crop gene-expression FASTA (InstaDeepAI 2024), TAIR10/Araport11 annotations, Ensembl Plants variation, and JASPAR 2024. File S1 contains the processed score tables used to generate the figures. Training and evaluation code, configs, and seeds are at <https://github.com/Cybing521/plant-bioinformatics-genes-paper>. Fine-tuned CNN and LoRA checkpoints are not deposited; they are reproduced from the public AgroNT-1B backbone and the scripts in that repository. Third-party FASTA and pretrained weights are obtained with the download commands in the repository, not re-hosted.

RESULTS

AgroNT outperforms the CNN on a unified Arabidopsis benchmark

On the held-out PGB test set ($n = 3,402$), AgroNT reached 86.8% accuracy (AUROC 0.939, AUPRC 0.935), compared with 75.4% accuracy (AUROC 0.842, AUPRC 0.835) for the CNN (Table 1; Figure 2). The AUROC difference of 0.097 had a bootstrap 95% CI of 0.086–0.108. McNemar’s test was $p = 6.7 \times 10^{-54}$; 532 genes were classified correctly by AgroNT and incorrectly by the CNN, versus 142 the reverse. Across three CNN seeds, test AUROC was 0.844 ± 0.003 and accuracy $76.3 \pm 0.9\%$, so the primary CNN result is not a single-run outlier. AgroNT is reported from seed 42. Single-tissue labels agreed with the tissue-averaged label for a mean of 92.1% of genes (bootstrap 95% CI 0.912–0.928 across 56 tissues; range 78.7–95.4%; median tissue–tissue Spearman 0.851; Figure S1).

Proximal flanks carry predictive signal; variant ranks are architecture-specific

On the structured flanking set, natural-variant $|\Delta|$ tracks follow the annotated compartments (Figure 3a,b). Matched mutagenesis ranked mean importance per base as 5’UTR (0.073) > terminator (0.066) > promoter (0.060) > 3’UTR (0.057) for the CNN and promoter (0.558) > 3’UTR (0.545) > 5’UTR (0.501) > terminator (0.483) for AgroNT (Figure 3d). Absolute mutagenesis scales are not comparable across architectures. DeepLift attributions used for motif discovery ranked 5’UTR (0.0088) > 3’UTR (0.0081) > promoter (0.0048) > terminator (0.0038) (Figure 3c). TF-MoDISco recovered four positive and four negative patterns; TOM-TOM against JASPAR 2024 mapped hits that include DOF-family factors (DOF1.5, DOF3.4/3.6, DOF5.1/5.8, CDF5), SPL8, NAC005, and AGL1

(Figure 3h). Because short DOF-like cores also appear among negative patterns, motif annotation is provisional.

CNN scoring of 27,662 natural flanking variants mapped median $|\Delta|$ as 5’UTR (0.00269; bootstrap 95% CI 0.00243–0.00291; $n = 2,542$) > 3’UTR (0.00112; 0.00099–0.00125; $n = 4,148$) > terminator (0.00058; 0.00055–0.00063; $n = 13,573$) > promoter (0.00043; 0.00040–0.00045; $n = 7,399$). Two-sided Mann–Whitney tests with Benjamini–Hochberg correction left 5’UTR above 3’UTR ($q = 4.6 \times 10^{-61}$; rank-biserial $r_{rb} = 0.24$). Promoter versus terminator was not significant after correction ($q = 0.19$). AgroNT median $|\Delta|$ on 26,518 overlapping sites ranked terminator (0.00202; 0.00196–0.00208) > 3’UTR (0.00118) > 5’UTR (0.00085) > promoter (0.00054) (Table 2; Figure 3e–g); terminator exceeded promoter at $q < 10^{-300}$ ($r_{rb} = -0.57$). Promoter was the least sensitive compartment in both maps. Attribution, mutagenesis, and variant ranks are not required to agree beyond that shared promoter-lowest pattern, and the two variant protocols are not allele-matched. An exploratory overlap of 17 reported eQTL variants had a higher CNN median $|\Delta|$ than the flanking background (0.00205 vs. 0.00065; Mann–Whitney $p = 0.041$; 10,000-permutation $p = 0.059$; Figure S2) and is not used as confirmatory evidence.

Arabidopsis ranking holds under crop PGB transfer

On the four crop PGB test sets, which were never used for Arabidopsis training, the Arabidopsis-trained CNN reached AUROC 0.716 (rice), 0.685 (maize), 0.763 (tomato), and 0.759 (soybean). The same Arabidopsis AgroNT LoRA checkpoint reached 0.815, 0.894, 0.838, and 0.878 (Figure 4a–c). AgroNT remained ahead on every crop; the largest gap was maize. Crop AUROC was lower than Arabidopsis self-performance (CNN 0.842; AgroNT 0.939). AgroNT pretraining included these crop genomes, so the crop readout is cross-species transfer of Arabidopsis adapters, not a species-unseen test. Retraining after midpoint-cropping the 6 kb windows to 3 kb or 1.5 kb gave CNN AUROC 0.542 and 0.545 and AgroNT AUROC 0.625 and 0.580 (Figure 4d). Under the TSS-anchored 5 kb+1 kb geometry, midpoint cropping removes most TSS-proximal sequence, so that series tests this window geometry rather than gene-body centering. The structured 3,020-bp flanking representation, which keeps complete annotated proximal flanks, reached CNN test AUROC 0.805 on its chromosomal split; that number is not pooled with the PGB truncation series because split and sequence layout differ.

DISCUSSION

The contest this study is built to win is a locked-protocol comparison: LoRA-tuned AgroNT versus a DeepSEA-style 1D-CNN on tissue-averaged Arabidopsis PGB labels. Under that protocol the AUROC gain is 0.097 (bootstrap 95% CI 0.086–0.108; McNemar $p = 6.7 \times 10^{-54}$). The direction agrees with AgroNT’s published PGB suite (Mendoza-Revilla *et al.* 2024). LoRA trains 1.6% of AgroNT parameters. When a ~ 0.1 AUROC improvement on this binary task is the decision criterion, the LoRA checkpoint is the stronger of the two models reported here; when a 2.7 M-parameter network is the operating constraint, the CNN is the compact baseline in the same table. The comparison is defined against this DeepSEA-style architecture, not against plant-specific flanking CNNs that can exceed 80% accuracy on other high/low datasets (Peleke *et al.* 2024), and not as a claim that foundation models are intrinsically stronger outside this protocol.

On the structured representation, mutagenesis and DeepLift both place predictive importance in proximal annotated flanks. These maps are exploratory with respect to architecture ranking: the CNN was retrained on concatenated flanks, AgroNT was not, and variant $|\Delta|$ used mapped alleles for the CNN versus three-alt means for AgroNT. Within those rules, CNN DeepLift and CNN mutagenesis agree that the 5’UTR is among the strongest compartments, consistent in direction with prior

Table 1 Unified benchmark on the PGB Arabidopsis held-out test set (primary seed 42; $n = 3,402$). Δ AUROC = 0.097 (bootstrap 95% CI 0.086–0.108). McNemar: 532 vs 142 discordant pairs, $p = 6.7 \times 10^{-54}$. CNN three-seed AUROC 0.844 ± 0.003 . AgroNT is one seed. Training time on one RTX 4090: 15 min (CNN) vs 240 min (AgroNT LoRA). Fine-tuned weights are not deposited.

Model	Trainable params	Accuracy	AUROC (95% CI)	AUPRC
1D-CNN (DeepSEA-style)	2.7 M	75.4%	0.842 (0.829–0.855)	0.835
AgroNT-1B (LoRA)	16.2 M of 985 M	86.8%	0.939 (0.931–0.947)	0.935

Table 2 Predicted $|\Delta|$ in high-expression probability by flanking region (250 genes). CNN: mapped alternate allele. AgroNT: mean over three non-reference bases. Absolute $|\Delta|$ is not compared across architectures because the allele protocol differs.

Region	n (CNN)	CNN median $ \Delta $	n (AgroNT)	AgroNT median $ \Delta $
5'UTR	2,542	0.00269	2,474	0.00085
3'UTR	4,148	0.00112	3,997	0.00118
Terminator	13,573	0.00058	12,958	0.00202
Promoter	7,399	0.00043	7,089	0.00054

reports that UTR sequence contributes to plant expression prediction (Peleke *et al.* 2024). Variant $|\Delta|$ ranks UTR > terminator > promoter for this CNN (5'UTR vs 3'UTR $q = 4.6 \times 10^{-61}$) and terminator > UTR > promoter for AgroNT. Promoter is the shared least-sensitive compartment. Attribution, mutagenesis, and variant maps are not a causal cis-regulatory atlas.

High tissue–tissue correlation and 92.1% mean label concordance (bootstrap 95% CI 0.912–0.928) support the averaged binary benchmark for architecture comparison. Concordance as low as 78.7% is the scope condition for organ-specific programs, which remain a tissue-conditioned task (Peleke *et al.* 2024). Crop PGB test sets keep the same model ranking (CNN AUROC 0.685–0.763; AgroNT 0.815–0.894). Those sets were not used in Arabidopsis training. The two crop readouts are still not the same experiment: the CNN never saw crop genomes, whereas AgroNT’s backbone already included those species in pretraining. Midpoint cropping of the published 6 kb window is tied to this TSS-anchored geometry for both architectures.

The evaluation is defined for tissue-averaged Arabidopsis PGB labels, a TSS-anchored 6 kb window, LoRA training of at most three epochs, and a single AgroNT seed. It is not the official 56-tissue PGB regression. DeepLift motif discovery was run on the CNN; AgroNT mutagenesis used the PGB LoRA checkpoint on concatenated flanks without retraining on that representation; variant $|\Delta|$ used mapped alleles for the CNN and three-alt means for AgroNT on 250 genes; the eQTL overlap used $n = 17$ associations and a permutation $p = 0.059$; structured UTR lengths are fixed at 500 nt. Fine-tuned weights are reproduced by retraining rather than downloaded. Tissue-of-expression labels, accessibility or methylation (O’Malley *et al.* 2016; Takou *et al.* 2025), larger eQTL sets, and longer-context DNA language models (Avsec *et al.* 2021) are outside this protocol.

CONCLUSIONS

On a derived Arabidopsis PGB task with tissue-averaged labels, LoRA-tuned AgroNT outperforms this DeepSEA-style 1D-CNN by a statistically supported margin under the reported budgets. Predictive importance concentrates in proximal annotated flanks. CNN variant $|\Delta|$ is led by the 5'UTR and AgroNT variant $|\Delta|$ by the terminator; both maps leave the promoter lowest under their stated allele protocols. Tissue-averaged labels align with most single-tissue binary axes. The same Arabidopsis checkpoints keep that ranking on rice, maize, tomato, and soybean PGB test sets as a cross-species transfer experiment.

DATA AVAILABILITY

Primary inputs are public: PGB Arabidopsis and crop gene-expression FASTA (InstaDeepAI 2024), TAIR10/Araport11 annotations, Ensembl Plants variation, and JASPAR 2024. File S1 contains processed score tables used for the figures. Training and evaluation code, configs, and seeds are at <https://github.com/Cybing521/plant-bioinformatics-genes-paper>. Fine-tuned CNN and LoRA checkpoints are not deposited; they are reproduced from the public AgroNT-1B backbone and the training scripts in that repository.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

AUTHOR CONTRIBUTIONS

To be completed before submission (CRedit). All authors will approve the final version and accept responsibility for accuracy.

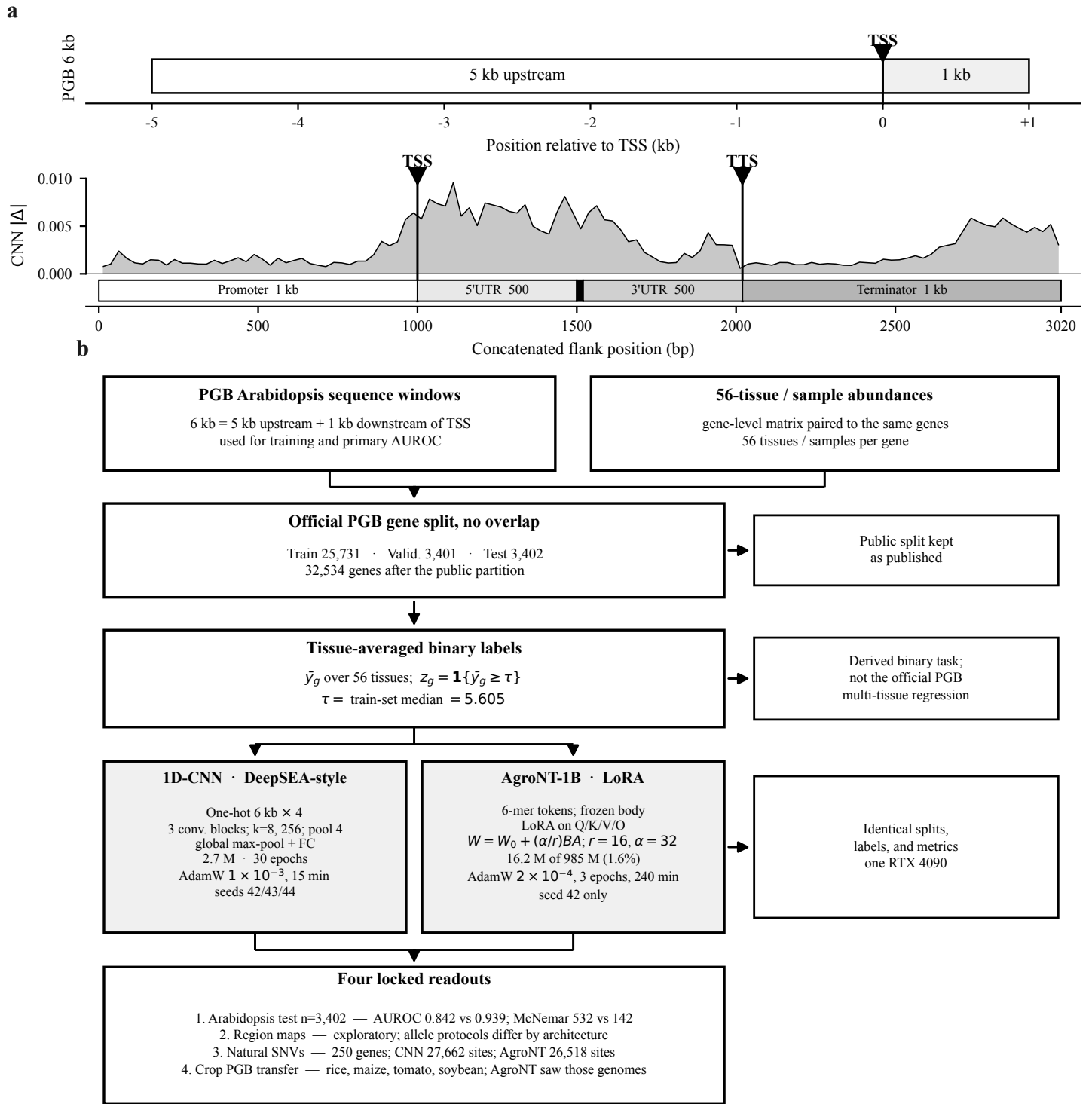


Figure 1 Locked evaluation protocol. (a) Sequence geometry. Top: each Plants Genomic Benchmark (PGB) Arabidopsis gene is a 6 kb transcription-start-site (TSS)-anchored window (5 kb upstream + 1 kb downstream). Bottom: a separate 3,020-bp concatenation of promoter, 5'UTR, 3'UTR, and terminator (20-nt N gap) used only for interpretability and variant scoring. The gray trace is mean CNN site-wise $|\Delta|$ in 25-bp bins (same scores as Figure 3a); TSS is the promoter/5'UTR junction and TTS is the 3'UTR/terminator junction. (b) Study flow. Binary labels collapse 56 tissue/sample abundances to a gene mean $\bar{y}_g$ and threshold at the training-set median $\tau = 5.605$ (equation 1). The official PGB gene split (25,731/3,401/3,402) is shared by a DeepSEA-style 1D-CNN (2.7 M parameters; seeds 42/43/44) and AgroNT-1B with LoRA on query/key/value/output projections (equation 2; seed 42). Crop PGB tests apply the Arabidopsis checkpoints without crop fine-tuning; AgroNT pretraining already included those genomes.

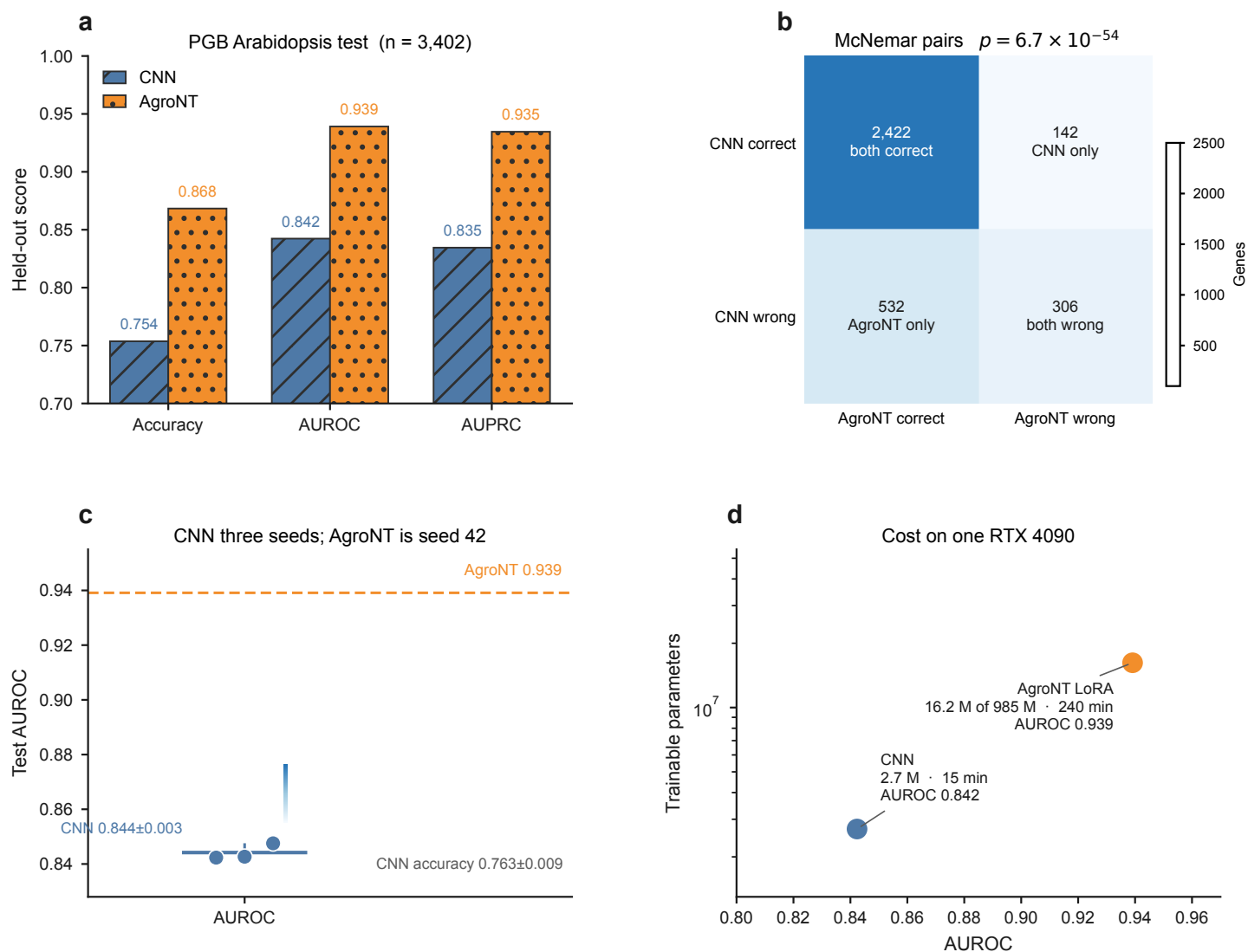


Figure 2 Head-to-head comparison on the PGB Arabidopsis test set. (a) Accuracy, AUROC, and AUPRC (Table 1). (b) McNemar 2×2 of paired calls ($n = 3,402$; equation 6). Off-diagonal cells are the 532 vs 142 discordant genes. (c) CNN three-seed AUROC and accuracy; the dashed line is the single AgroNT seed (42). (d) AUROC versus trainable-parameter count on one RTX 4090. AgroNT is plotted at 16.2 M trainable parameters.

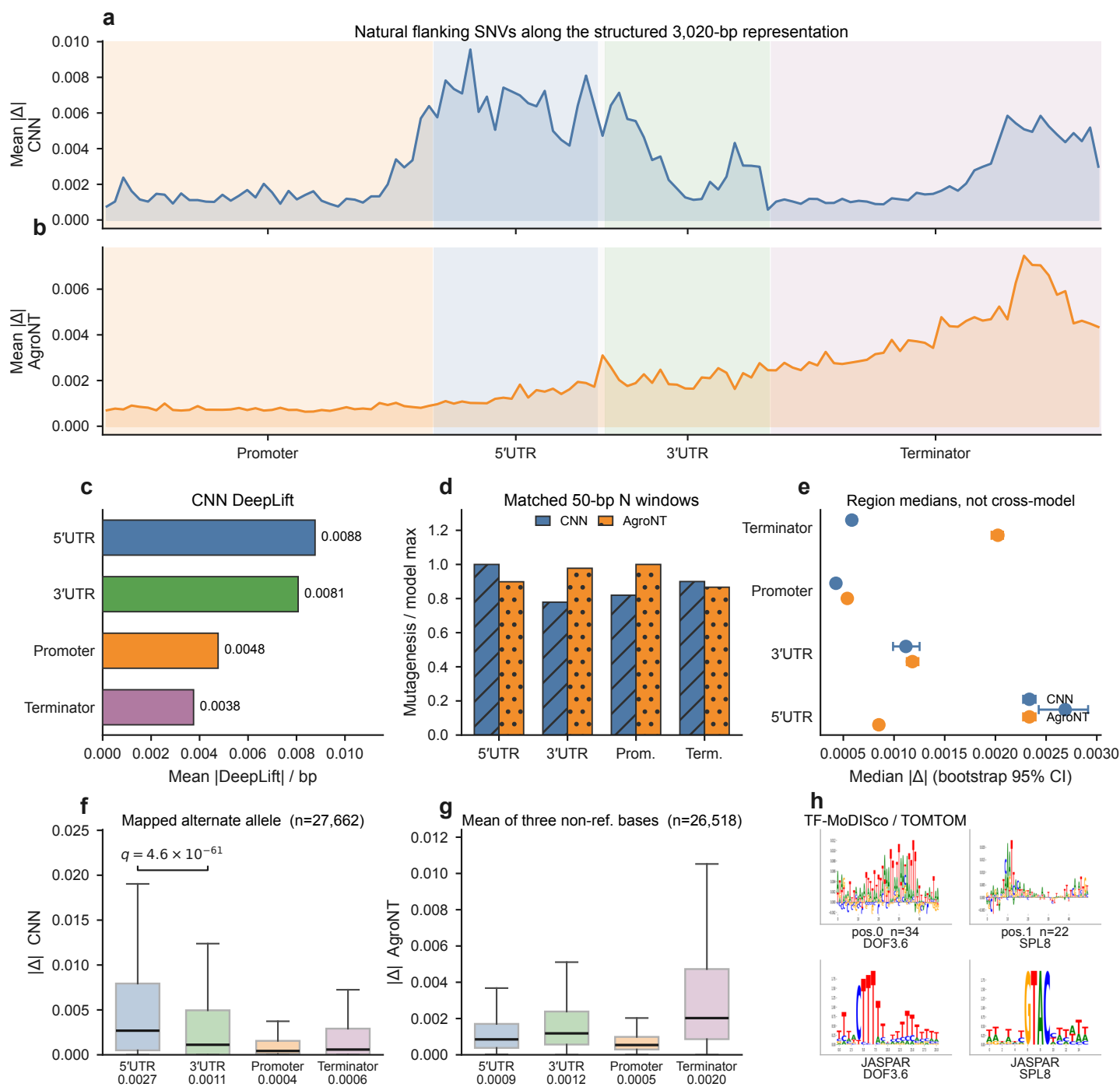


Figure 3 Region importance and variant effects on the structured flanking representation (equation 5). (a,b) Mean $|\Delta|$ in 25-bp bins along the 3,020-bp concatenation for the CNN and AgroNT. Shading marks promoter, 5'UTR, 3'UTR, and terminator. (c) Mean absolute DeepLift attribution per base (CNN). (d) Sliding-window mutagenesis (equation 4), normalized to each model's maximum; the CNN was retrained on this representation and AgroNT was not. (e) Regional median $|\Delta|$ with 10,000-resample bootstrap 95% CIs (within-model only). (f) CNN variant $|\Delta|$ by region (mapped alternate allele; 5'UTR vs 3'UTR Benjamini–Hochberg $q = 4.6 \times 10^{-61}$). (g) AgroNT $|\Delta|$ at overlapping sites (mean over three non-reference bases). (h) TF-MoDISco positive-pattern CWMs and JASPAR 2024 matches (DOF3.6, SPL8). Absolute $|\Delta|$ is not compared across architectures because the allele protocol differs.

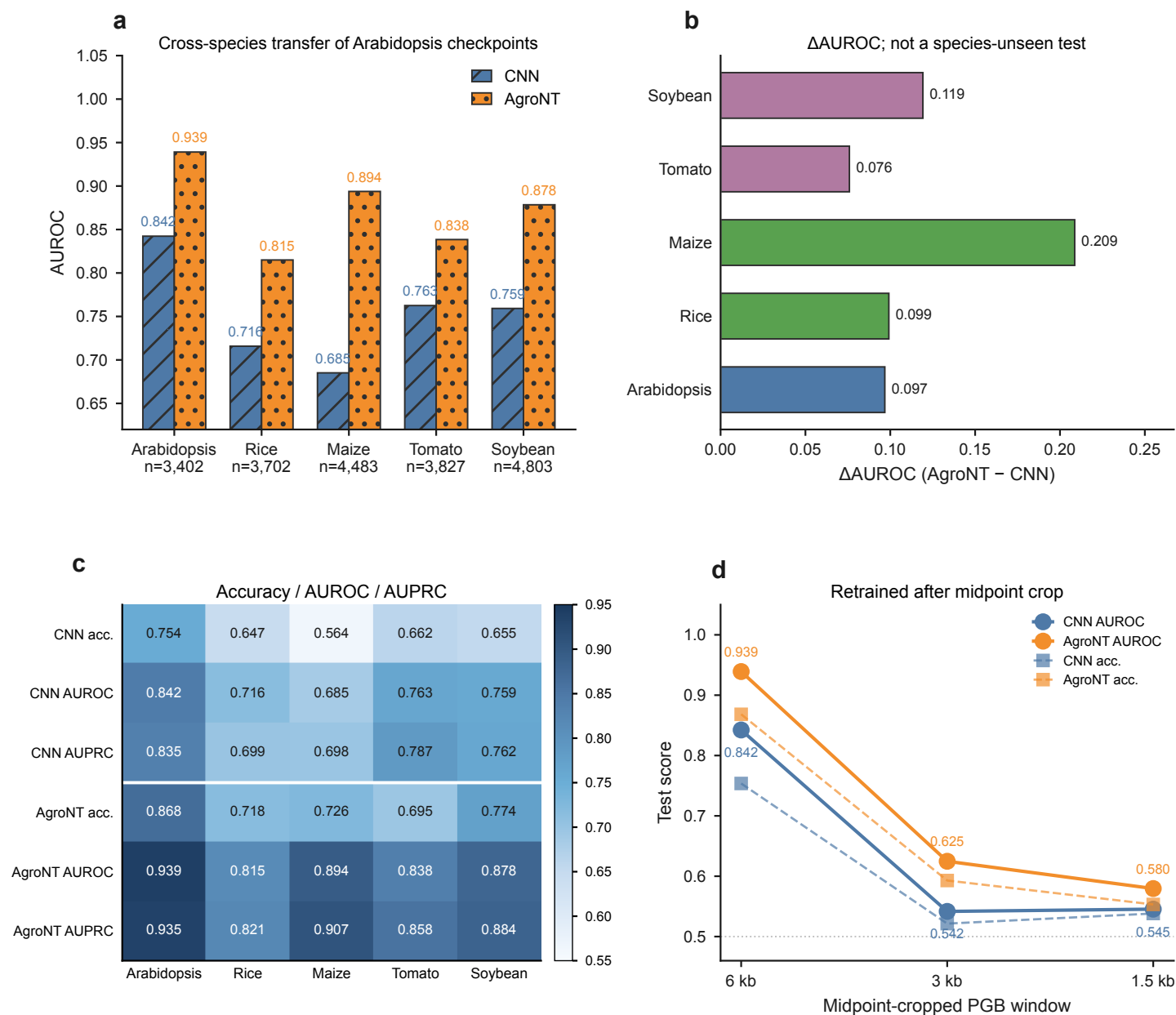


Figure 4 Cross-species transfer of Arabidopsis checkpoints and window geometry. (a) Arabidopsis-trained CNN and AgroNT LoRA checkpoints applied, without crop fine-tuning, to rice, maize, tomato, and soybean PGB test sets under each species' training-set median. Arabidopsis self-test is shown for reference. AgroNT pretraining included these crop genomes. (b) Δ AUROC (AgroNT - CNN) on the same five tests. (c) Accuracy, AUROC, and AUPRC for both models. (d) Test AUROC (solid) and accuracy (dashed) after retraining both models on midpoint-cropped 6 kb, 3 kb, and 1.5 kb PGB windows. The dotted line marks 0.5.

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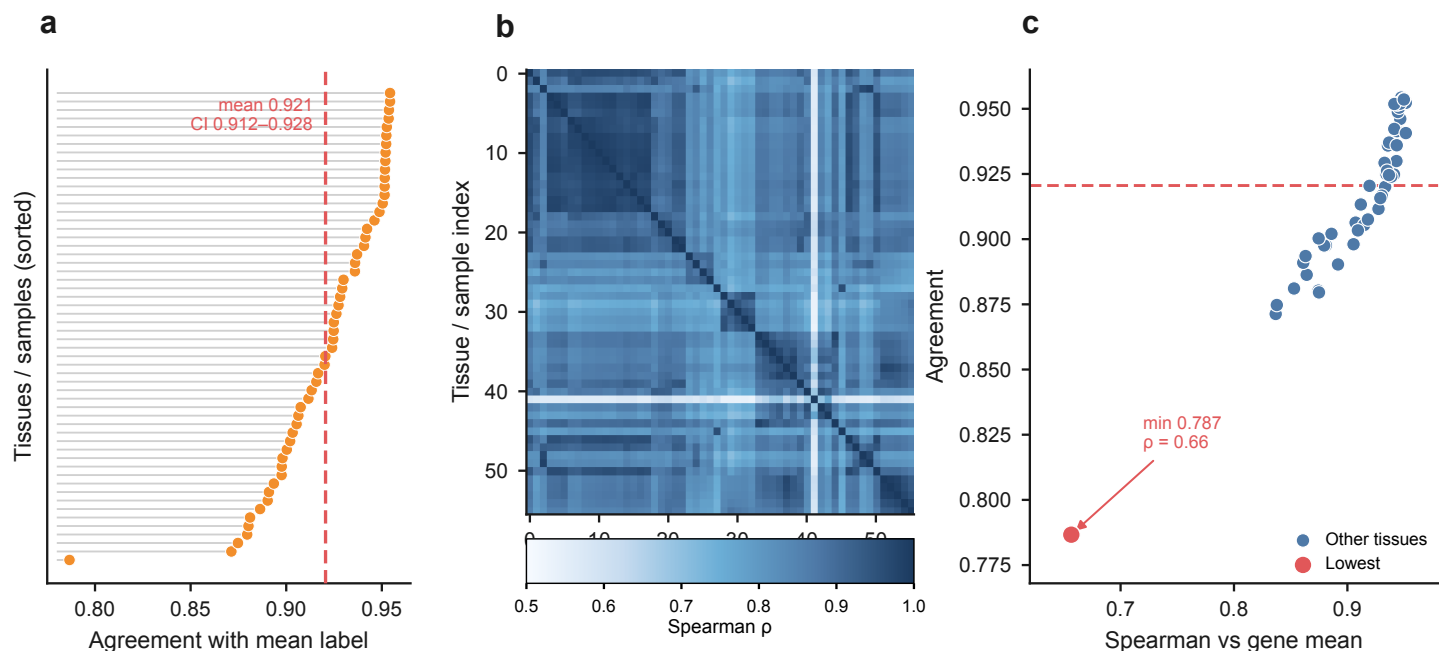


Figure S1 Scope of tissue-averaged labels across 32,534 genes and 56 tissues/samples. (a) Concordance of each tissue's median-based high/low label with the tissue-averaged label (mean agreement 0.921; bootstrap 95% CI 0.912–0.928 by resampling tissues; range 0.787–0.954). (b) Pairwise Spearman correlation among tissues (median $\rho = 0.851$). (c) Per-tissue agreement versus Spearman correlation with the gene-mean abundance. Cross-tissue CV: median 0.340, 75th percentile 1.37. File S1 contains the processed tables underlying this figure and the main tables.

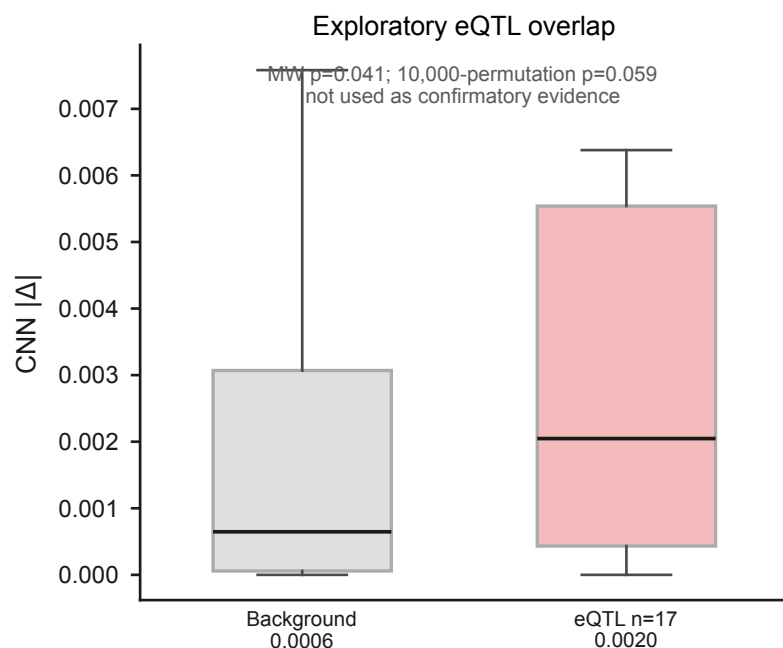


Figure S2 Exploratory overlap of 17 mapped eQTL associations with CNN flanking-variant $|\Delta|$. Median $|\Delta|$ was 0.00205 for eQTL sites versus 0.00065 for the background (one-sided Mann–Whitney $p = 0.041$; 10,000-permutation $p = 0.059$). The comparison is underpowered and is not used as confirmatory evidence.